Nearby Jobs

Vendor app and customer app changes for H3 dispatch

Jobs now reach the vendors **nearest** to the customer first, like Uber. This guide lists what the **vendor app** and the **customer app** need to change. Sign-in, tokens and the rest of the job flow (start code, completion code, photos, complete) stay the same.

Base URL	`https://home-services-backend-7cyx.onrender.com/api/v1`
Live reference	`https://home-services-backend-7cyx.onrender.com/api-docs` (sections "Vendor app - going online (nearby jobs)" and "Customer app - vendors nearby")
Version	21 October 2026

Summary of changes

App	Change	Effort
Vendor	Online / Offline switch + location every 30 s while online	New
Vendor	Poll available jobs every 5 s, show offers with a 60 s countdown	Update
Vendor	Decline button on offered jobs	New
Vendor	Handle "job no longer available" on Accept	Update
Customer	Optional: "6 vendors nearby, ~12 min" before booking	New
Customer	Send the address latitude / longitude with the booking	Check

1. How a job reaches a vendor

The map is split into H3 hexagons. When a customer books:

Step	What happens
1	The booking is placed in the hexagon of its address (needs `latitude` / `longitude` on the address).
2	Wave 1: the **5 nearest** online vendors within ~1 km who offer the service and are not busy get an offer lasting **60 seconds**.
3	Nobody accepts or everyone declines: wave 2 goes to vendors in a wider ring (~2.5 km), then ~5, ~8, ~12 km.
4	The **first vendor to accept** gets the job; every other offer is withdrawn.
5	Nobody nearby takes it, or the address has no location: the job is **opened to all** approved vendors.

A vendor only receives offers while **online** and while the app keeps sending its position. No position for 5 minutes means no offers; after 15 minutes the server sets the vendor offline.

2. Vendor app: Online / Offline switch

Add a switch on the home screen. It needs location permission (foreground is enough while the app is open).

GO ONLINE (AND EVERY 30 SECONDS WHILE ONLINE)

```
PUT /vendor/presence
Authorization: Bearer <accessToken>
Content-Type: application/json

{ "online": true, "latitude": 19.0636, "longitude": 72.8300 }
```

RESPONSE

```
{
  "success": true,
  "data": {
    "online": true,
    "latitude": 19.0636,
    "longitude": 72.83,
    "h3Cell": "88608b0b13fffff",
    "lastSeenAt": "2026-10-21T10:00:00.000Z",
    "onlineSince": "2026-10-21T09:30:00.000Z",
    "canReceiveOffers": true,
    "activeOffers": 1,
    "updateEverySeconds": 30
  }
}
```

GO OFFLINE

```
PUT /vendor/presence

{ "online": false }
```

- On app start call `GET /vendor/presence` (same response) to set the switch.
- While online, send the call again every `updateEverySeconds` (30) with the current position. Stop the timer when the vendor goes offline or logs out.
- Going offline hands back any offers not answered yet.
- `canReceiveOffers: false` while online means the vendor is not approved yet: show "Your profile is under review".
- **400** "Send your location" if going online without a position and none was sent before: ask for location permission.

- Call `{ "online": false }` on logout.

3. Vendor app: new jobs list

Keep using `GET /vendor/jobs/available`, but poll it **every 5 seconds** while the vendor is online and on that screen (and when a `JOB_OFFER` push arrives, once push is live). Each job now has three new fields:

ONE ITEM IN data.items

```
{
  "id": "caaae246-ba43-4642-9ce5-b3a8cfc0e332",
  "bookingNumber": "BK-000943",
  "service": { "id": "...", "name": "Electrician", "tier": "REPAIR" },
  "scheduledAt": "2026-10-21T04:59:00.000Z",
  "description": "Fan not working",
  "price": 299,
  "paymentMode": "PAY_AFTER_SERVICE",
  "area": { "city": { "name": "Mumbai" }, "zone": { "name": "Bandra West" }, "pincode": "400
    050" },
  "distanceKm": 0.23,
  "offer": { "expiresAt": "2026-10-21T10:01:00.000Z", "wave": 1 },
  "openToAll": false,
  "createdAt": "2026-10-21T10:00:00.000Z"
}
```

Field	Show
`distanceKm`	"0.2 km away" (null if either location is missing)
`offer`	Not null = offered to **this** vendor. Show "Offered to you - 52s left", counting down to `offer.expiresAt`, with **Accept** and **Decline**. Put these at the top.
`openToAll`	true = anyone may take it. Show "Open to all vendors" with **Accept** only.

When the countdown hits zero, remove the job from the list (or refresh); it may come back later as open to all.

`GET /vendor/jobs/available/{id}` returns the same fields for one job.

4. Vendor app: Accept and Decline

Action	Call	Result
Accept	`POST /vendor/jobs/{id}/accept`	Unchanged: the job becomes ASSIGNED to this vendor
Decline	`POST /vendor/jobs/{id}/decline` (no body)	`{ "declined": true }`; remove it from the list

409 on Accept ("This job is no longer available" / "Another vendor just took this job"): the offer ran out or someone else was faster. Show a short message and refresh the list.

409 on Decline: the offer had already run out. Just remove the job.

5. Customer app: vendors nearby (optional)

On the service or address screen before booking, show how many vendors are around.

REQUEST

```
GET /customer/nearby-vendors?latitude=19.0622&longitude=72.8315&categoryId=<service id>
```

RESPONSE

```
{
  "success": true,
  "data": {
    "vendorsNearby": 6,
    "nearestKm": 0.8,
    "etaMinutes": 12,
    "radiusKm": 5,
    "areas": [ { "cell": "87608b0b1ffffff", "center": [19.061, 72.832],
                 "boundary": [[19.07, 72.82], ...], "vendors": 4 } ]
  }
}
```

- Show "6 vendors nearby - about 12 min". With `vendorsNearby: 0`, show "Vendors are busy right now - we'll find one for you"; the booking can still be placed.
- `areas` is optional, for drawing shaded hexagons on a map. Exact vendor positions are never returned.

Booking location. Nearby matching uses the address's `latitude` and `longitude`. Make sure the address screen saves them (map pin or GPS) when creating or editing an address. Without them the job still works but goes straight to all vendors.

6. Push notifications (coming)

The server already sends a "New job near you" push with `data: { "type": "JOB_OFFER", "bookingId": "..." }`, but it only goes out once Firebase (FCM) is set up and device registration is added. Until then, polling every 5 seconds is how offers reach the vendor. Plan for the push to open the new jobs screen and refresh it.

7. Checklist

	App	Item
[ ]	Vendor	Online / Offline switch calling `PUT /vendor/presence`; initial state from `GET /vendor/presence`
[ ]	Vendor	Location permission; send position every 30 s while online; stop when offline
[ ]	Vendor	Go offline on logout
[ ]	Vendor	Poll `GET /vendor/jobs/available` every 5 s while online
[ ]	Vendor	Show `distanceKm`, the offer countdown to `offer.expiresAt`, and "Open to all"
[ ]	Vendor	Decline button: `POST /vendor/jobs/{id}/decline`
[ ]	Vendor	Handle 409 on Accept / Decline by refreshing the list
[ ]	Customer	Save latitude / longitude on addresses
[ ]	Customer	Optional: vendors nearby via `GET /customer/nearby-vendors`